Task 2: Experimental Terrain Visualization

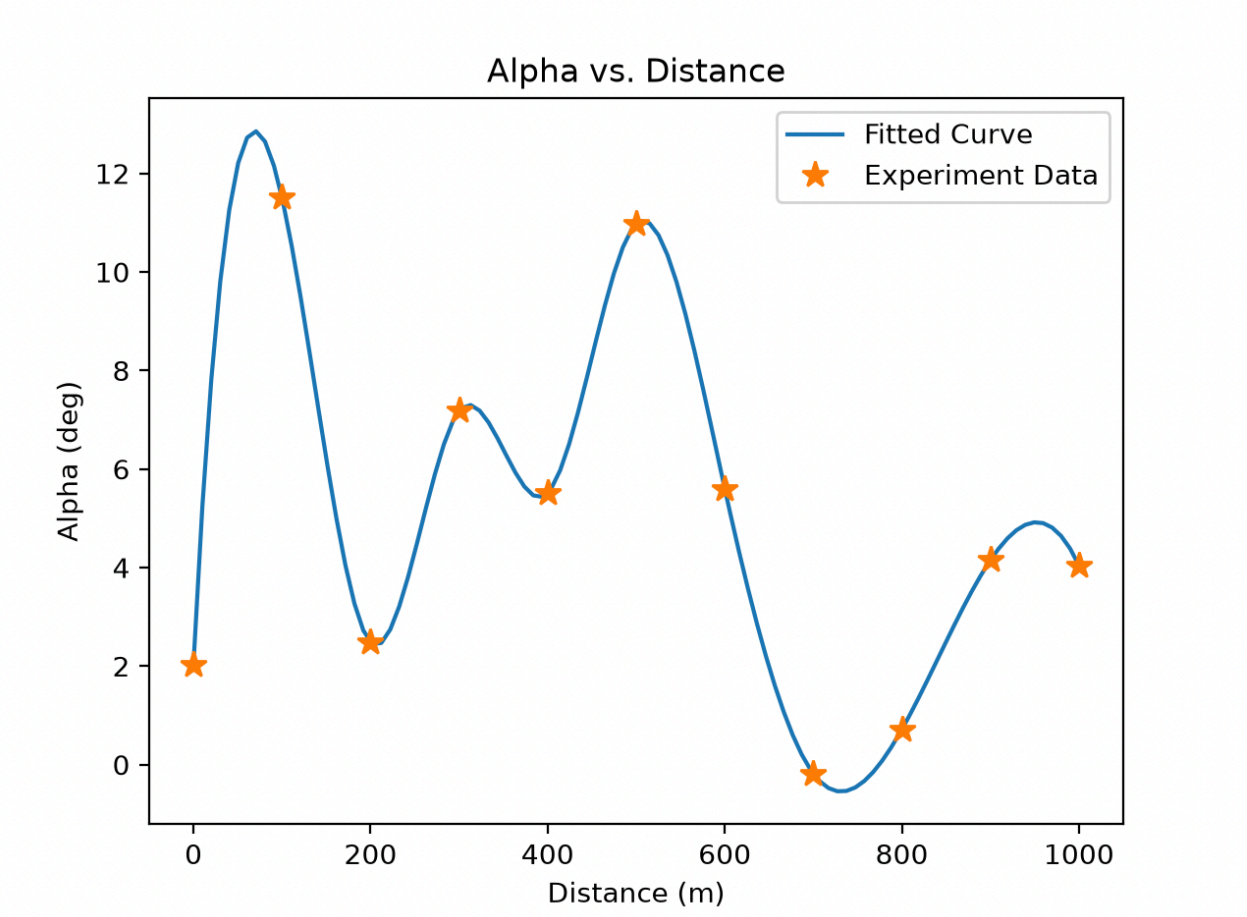


Figure 1: Terrain inclination angle $\alpha(x)$ *along the 1000 m experimental trajectory.*

Written Description & Terrain Analysis

Please include a copy of your figure and an explanation of what you are observing in your write-up. For the explanation portion, discuss if the curve is smooth or not, why it is so, and whether the curve goes through all the known data points.

The Angle versus Distance graph displays the terrain angle at various distances on the rover's path. The curve is smooth because the terrain angle does not suddenly drop or rise on the surface of mars. The created curve goes through all the data points and accurately reflects the terrain angle along the rover's path.

Task 5: Visualizing Motor Efficiency

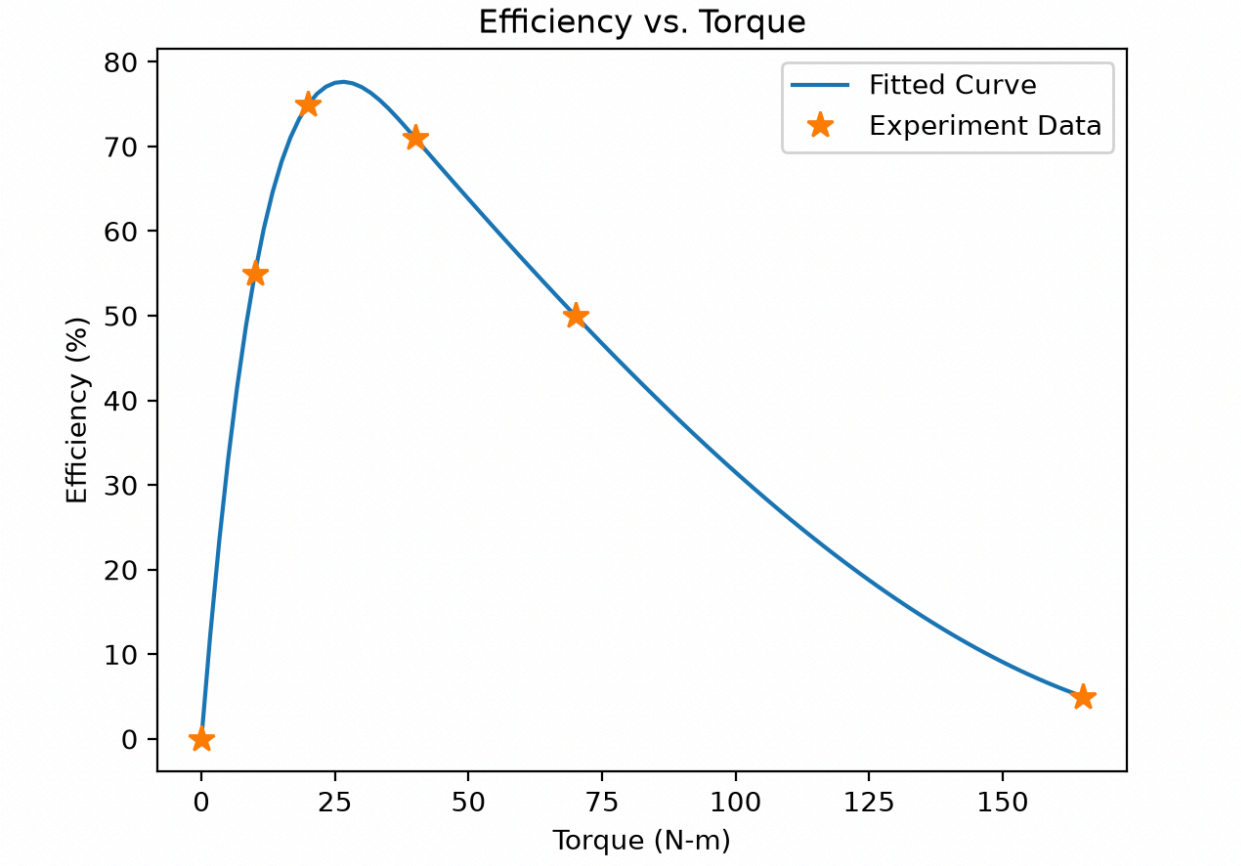


Figure 2: DC motor efficiency (η) as a function of motor shaft torque (τ)

Discussion of Efficiency Characteristics

The motor operates as its maximum efficiency when the torque value is at 26, shown by the maximum efficiency value on Figure 2. At low torques near no-load speed, fixed internal losses take up nearly all of the energy input. At high torques near stall torque, a lot of electrical power is lost due to resistors, causing efficiency to drop. Compared to Figure 2, this curve has a clear absolute maximum and a predictable path. Since efficiency is not dependent on random features, including the surface of Mars, its projection is more predictable. The curve is still smooth and goes through the knoy+wn data points.

Task 8: Rover Simulation & Dynamical Trajectory Analysis

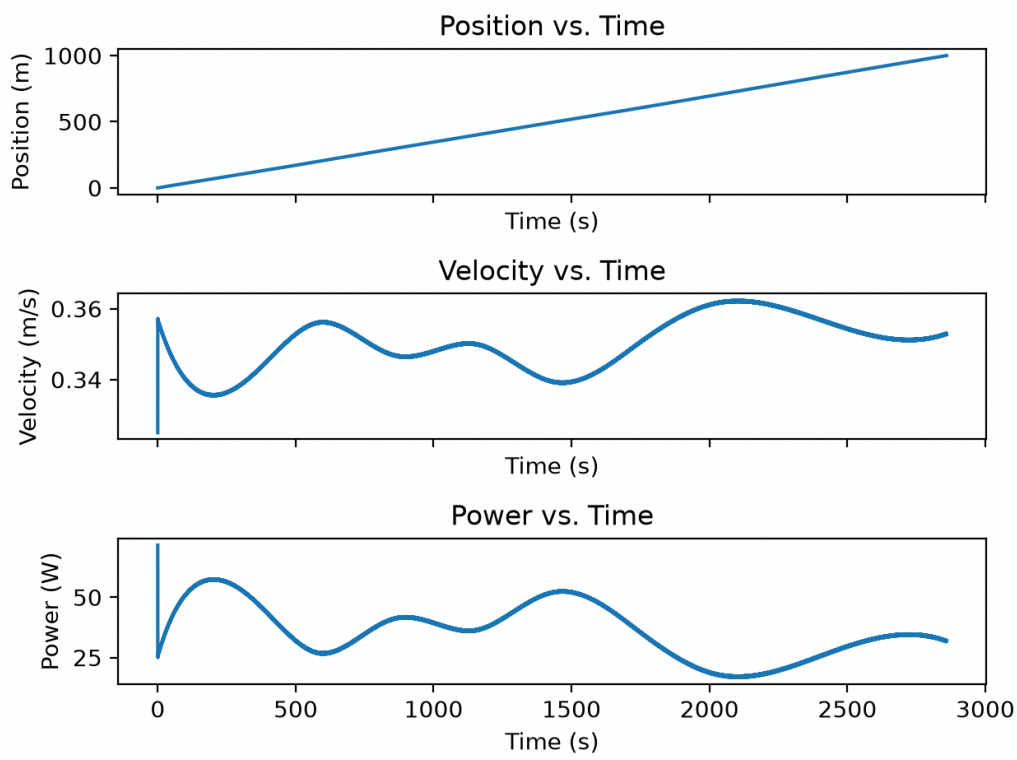


Figure 3: Time history of rover position, translational velocity, and instantaneous motor power over the experimental mission trajectory.

Discussion & Physical Interpretations

The position curve appears to be linear overall, while the velocity and power functions are nonlinear. When the rover travels near its terminal velocity for long stretches, position appears to be linear. The terrain slopes seen in Figure 2 cause small accelerations and decelerations. In the velocity graph, the points range from around 0.34 m/s to 0.36 m/s. Since position is the integral of velocity, it is inherently non-linear, even though the slopes vary so little that the graph appears linear.

From the plots, velocity is shown to be inversely related with power. Power is obtained by multiplying the torque and angular velocity. On flat or downhill terrain where the rover gets to its highest speeds, the torque drops towards zero, resulting in less power drawn. On inclines, the lower velocities require more motor torque, and therefore high power draw.

The velocity profile is directly related to the plot described in Figure 1. On uphill slopes, gravity goes against the rover's motion, decreasing acceleration and causing a speed drop, and the inverse is true on downhill slopes.

The velocity graph is only not smooth at the beginning, when the rover's motion is first documented. Even though it may look like a jump, it simply is a difference in the initial conditions and the rover's steady speed. Apart from this, the rest of the velocity curve is smooth. The mass of the rover provides mechanical inertia, preventing sudden velocity changes even if the terrain slope abruptly changes.

Task 9: Analysis of Energy Needs

Your boss has informed you that the rover will be equipped with a 0.9072e6 [J] Lithium Iron Phosphate battery back. Can the rover complete the case defined by experiment1 (in define_experiment.py) with this

battery pack? Please include your answer to this question and how you arrived at it in your write-up. Note that this task is a continuation of Task 8.

Summary of Telemetry Metrics

- Total Distance Traveled (d_{total}): 1000 m
- Total Mission Time ($t_{completion}$): 2859 s
- Average Velocity (v_{avg}) & Max Velocity (v_{max}): 0.350 m/s
- Total Battery Energy Extracted ($E_{battery}$): 1.128e6 J
- Energy per Unit Distance (E_{dist}): 1,128.2 J/m

The battery pack provides 0.9072e6 Joules of energy. In order to see if the rover can complete the mission with this amount of energy, we can take the integral of the power function. Approximating the integral using trapezoidal approximation. Through its journey, the rover would have to expend around 1.12e6 Joules to complete the mission. This is above the energy provided by the battery pack, meaning that the rover will be unable to complete the mission given only this energy source.